CENTRAL BOARD OF SECONDARY EDUCATION

CLASS 10 MATHEMATICS UNIT TEST - QUADRATIC EQUATIONS

CLASS 10 — MATHEMATICS

Time Allowed: 1 Hours **Maximum Marks:** 29

General Instructions:

1. This question paper contains 5 Sections: A, B, C, D, and E.
2. **Section A** comprises Multiple Choice Questions (MCQs) and Assertion-Reason questions of 1 mark each.
3. **Section B** comprises Very Short Answer (VSA) type questions of 2 marks each.
4. **Section C** comprises Short Answer (SA) type questions of 3 marks each.
5. **Section D** comprises Long Answer (LA) type questions of 5 marks each.
6. **Section E** comprises Case Based integrated units of assessment of 4 marks each with sub-parts.
7. All Questions are compulsory. Internal choice is provided in some questions.
8. Use of calculators or electronic devices is not permitted.

SECTION A (Multiple Choice & Objective Questions - 1 Mark each)

Q1. Assertion (A): The equation **[1]**

$$x^2 + 3x + 4 = 0$$

has no real roots. Reason (R): For a quadratic equation

$$ax^2 + bx + c = 0$$

, if

$$b^2 - 4ac < 0$$

, the roots are imaginary.

(A) Both (A) and (R) are true and (R) is the correct explanation of (A)
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A)
(C) (A) is true but (R) is false
(D) (A) is false but (R) is true

Q2. The exponent of **[1]**

$$2$$

in the prime factorisation of

$$144$$

is:

(A) 4
(B) 5
(C) 6
(D) 3

Q3. If the discriminant of equation **[1]**

$$ax^2 + bx + c = 0$$

is zero, then the roots are:

(A) Real and equal
(B) Real and distinct
(C) Imaginary
(D) Zero

Q4. If the equation **[1]**

$$x^2 - kx + 9 = 0$$

has equal roots, then the value of

$$k$$

is:

(A) $+-6$
(B) 6
(C) -6
(D) $+-3$

Q5. The sum of the reciprocals of Rehman's ages (in years) **[3]**

$$3 \text{ years ago}$$

and

$$5 \text{ years from now}$$

is

$$\frac{1}{3}$$

. Find his present age.

SECTION B (Very Short Answer Questions - 2 Marks each)

Q6. Find the values of **[2]**

$$k$$

for which the quadratic equation

$$kx(x - 2) + 6 = 0$$

has two equal real roots.

Q7. A motor boat whose speed is **[3]**

$$18 \text{ km/h}$$

in still water takes

$$1 \text{ hour}$$

more to go

$$24 \text{ km}$$

upstream than to return downstream to the same spot. Find the speed of the stream.

Q8. Solve for **[3]**

$$x$$

by quadratic formula:

$$4sqrt(3)x^2 + 5x - 2sqrt(3) = 0$$

.

SECTION C (Short Answer Questions - 3 Marks each)

Q9. Two water taps together can fill a tank in **[5]**

$$9\frac{3}{8}\ \text{hours} = \frac{75}{8}\ \text{hours}$$

. The tap of larger diameter takes

$$10\ \text{hours}$$

less than the smaller one to fill the tank separately. Find the time in which each tap can separately fill the tank.

Q10. A rectangular park of length **[4]**

$$L$$

and breadth

$$B$$

has an area of

$$528\ \text{m}^2$$

. The length of the plot is one more than twice its breadth. (i) Formulate the quadratic equation representing this situation. [1 Mark] (ii) Find the length and breadth of the park by factoring. [2 Marks] (iii) Calculate the cost of fencing the park at

$$\text{Rs. }\ 25\ \text{per meter}$$

. [1 Mark]

SECTION D (Long Answer Questions - 5 Marks each)

Q11. An express train takes **[5]**

$$1\ \text{hour}$$

less than a passenger train to travel

$$132\ \text{km}$$

between Mysore and Bangalore. If the average speed of the express train is

$$11\ \text{km/h}$$

more than that of the passenger train, find the average speed of the two trains.

END OF QUESTION PAPER